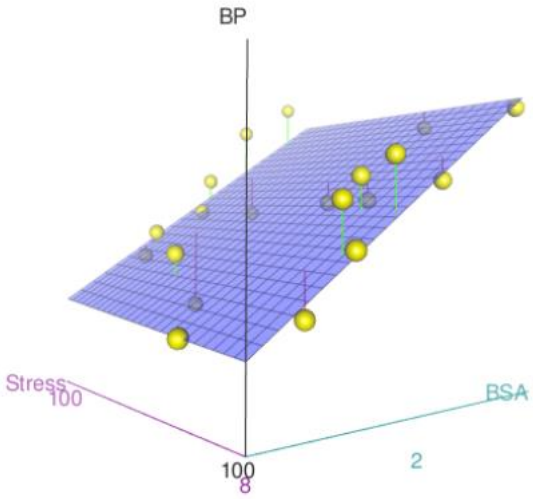
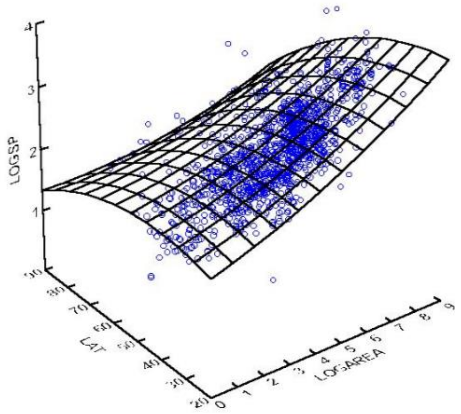
Chapter 6.5: Extra reading material

Multiple linear regression



Nonlinear multiple linear regression



Given that

$$\hat{y_i} = \hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \ldots + \hat{\beta}_k x_{i,k}$$

Interpretation of the Model Parameters

- Each β coefficient represents the change in the mean response, E(y), per unit increase in the associated predictor variable when all the other predictors are held constant.
- For example, β_1 represents the change in the mean response, E(y), per unit increase in x_1 when x_2, x_3, ..., x_k are held constant.
- The intercept term, β_0, represents the mean response, E(y), when all the predictors x_1, x_2, ..., x_k, are all zero (which may or may not have any practical meaning).

Example:

$$Y = 42 + 2.3X_1 + 11I(X_2)$$

The above MLR model shows the predicted model of the height (in cm) of a specific plant (Y) based on the amount of bacteria in the soil (X_1) and whether the plant is located in partial or full sunlight (X_2). X_2 is an indicator function, $I(X_2)=0$ if the plant is in partial sunlight, $I(X_2)=1$ if the plant is in full sunlight. Bacteria is measured in 1000 per ml of soil.

Interpreting coefficients of continuous explanatory variable (X_1)

As long as plants were in the same type of sunlight, an increase of 1000 per ml bacteria count of soil, plants are expected to grow by 2.3 cm on the average.

Interpreting coefficients of categorical explanatory variable (X_2)

The predicted height of the plant is expected to increase by 11 cm compared to when it is in partial sunlight, assuming the amount of bacteria in the soil remain constant.

Interpreting intercept

We would expect an average height of 42 cm for plants in partial sunlight with no bacteria in the soil. However, this is only a meaningful interpretation if it is reasonable that both X_1 and X_2 can be 0, and if the data set actually included values for X_1 and X_2 that were near 0.

If neither of these conditions are true, then β_0 really has no meaningful interpretation. It just anchors the regression line in the right place. In our case, it is easy to see that X_2 sometimes is 0, but if X_1, the bacteria level, never comes close to 0, then the intercept has no real interpretation.

Remark:

Interpreting coefficients when explanatory variables are correlated

Don't forget that each coefficient is influenced by the other variables in a regression model. Because predictor variables are nearly always associated, two or more variables may explain some of the same variation in Y.

Therefore, each coefficient does not measure the total effect on Y of its corresponding variable.

How to write a fitted model with both continuous and categorical explanatory variables?

1.

Fitted model : (No interaction)

$$\hat{y} = \begin{cases} \hat{\beta}_0 + \hat{\beta}_1 x & \text{if type} = 1 \\ (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 x & \text{if type} = 2 \end{cases}$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 \, I\{\text{type} = 2\}$$

Indicator function

type 1 : 'baseline', or 'reference' level

type 2 : additional effect

In R: lm(y ~ x + type)

2.

Fitted model : (With interaction effect)

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 \, I\{\text{type} = 2\} + \hat{\beta}_3 x \, I\{\text{type} = 2\}$$

$$= \begin{cases} \hat{\beta}_0 + \hat{\beta}_1 x & \text{if type} = 1 \\ (\hat{\beta}_0 + \hat{\beta}_2) + (\hat{\beta}_1 + \hat{\beta}_3) x & \text{if type} = 2 \end{cases}$$

In R: lm(y ~ x * type)